

Daily Research report

Subject:

Implementation of a runtime for
executing computations on
volunteer machines in a
resource-limited environment

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Tasks Achieved:

- Getting Started with the Rust Programming Language
- Reading the Grok article and testing some namespaces working mechanism (end with users, cgroups and times) and global summary.

Why I chose Rust as the programming language to implement my runtime

Rust is one of the most most used languages for OS programming.

For choosing the right programming language to use here, i did a little comparaisn of the languages i can use, C, bash only, Go and Rust and saw that

- **Bash:** Provides a lightweigh, easy, universal and direct access to sys tools (such as sys calls and commands), is portable and easy to debug, but the error handling there is a little difficult because of the static aspect of the code there. it doesn't have complete libraries or data structures, what makes it less versatile and not worthy to be used for a big project.
- **The C programming language:** It also provides easy ways to access to system tools and calls as it is a low level language, it has well defined libraries and is used by many great companies such as Docker, runc containers... and is the native language of many Os (linux, windows, ...) But compared to bash, it can be a little slow, is also difficult to maintain and doesn't provide an integrated memory management tool (which is principally what we need to implement our runtime)
- **Rust programming language:** As said before Rust is one of the most most used languages for OS programming. It has been created by a mozilla developer Graden Hoare, from the need to avoid memory errors in a program, managing concurrency and being very performant.
Despite it is a little difficult to understand and use, because of its borrow checker tool for memory management, and because ressources on namespaces are a little rare in rust, it provides a reliable security and memory handling; It also have well-supplied data structures and has some framework adapted to the sys programming (such as nix, caps, clap etc ...). in addition to having performances equivalent to those of the C Language.
- **Go programming language:** It has more supplied libraries than C, has a garbage

collector for memory management but is slower than C and Rust, provides less low level than them because it is more adapted to cloud programming. Its first goal is to be simple for prototyping, and efficient to be the most used for cloud computing.

some other advantages of Rust are

- It is already multiplatform (principally windows and linux)
- Doesn't need to install a heavy environment to use or to execute its programs.
- Supports Object oriented programming despite it is a low level programming language such as C
- Rust is a language that has very strong memory protections which tends to prevent many errors common in C (segmentation errors, memory allocation...)
- Embedded or low-level applications

Ref: [Medium article](#)
[code crafter](#)